

EXECUTIVE BRIEFING

CLINICAL OPERATIONS & BUSINESS INTELLIGENCE BRIEF

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BUSINESS CONTEXT

Over the past 24 months, leadership at Zion Tech Hub Facility has become increasingly concerned about the hospital's ability to efficiently manage patient flow across the inpatient system.

Despite continued patient activity, several departments have reported challenges in moving patients through the inpatient system. This operational friction compromises service excellence.

KEY CONCERNS UNDER REVIEW:

- Operational gridlock within high-volume clinical departments.
- Adverse impacts of delayed discharge on overall patient outcomes.
- Compounding revenue leakages affecting hospital financial sustainability.

DATA STRUCTURE & ANALYTICAL APPROACH

The analysis integrated approximately **1 million relational records** across core clinical operations, finance, demographics, and billing.

RELATIONAL DATA MATRIX

Unified analytical dataset integrated using:

Encounter ID

(Encounter activities)

Patient ID

(Demographics & diagnoses)

Payer ID

(Financial transactions & plans)

KEY OPERATIONAL METRICS

Inpatient Volume:

Admissions and tracking.

ALOS:

Average Length of Stay.

Total Bed Day Burden:

Ward utilization.

Readmission Rate:

Core clinical quality metric.

Data preparation was executed in Excel, analytical querying modeling in PostgreSQL, and final visualization structures built in Tableau.

Operation Analytics Dashboard

Encounte..

2023

▼

ALOS Year

2023

▼

Average LOS

4.7

Inpatient Volume

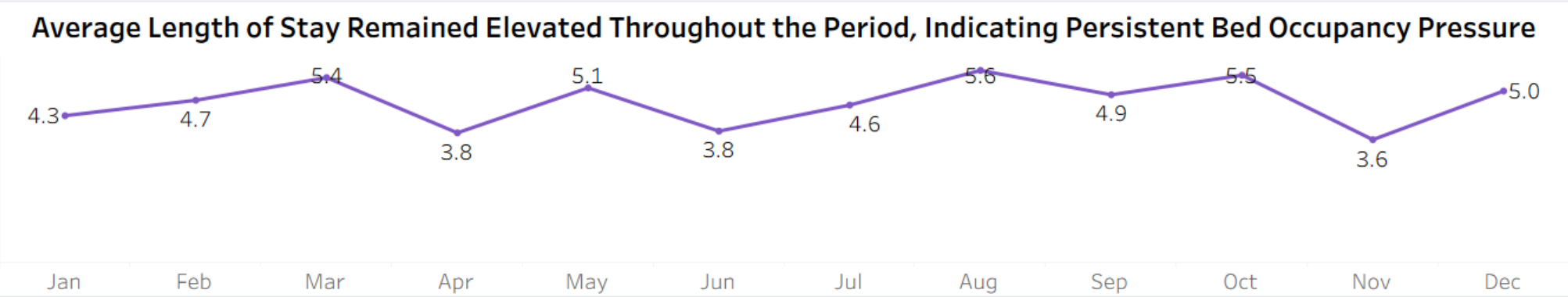
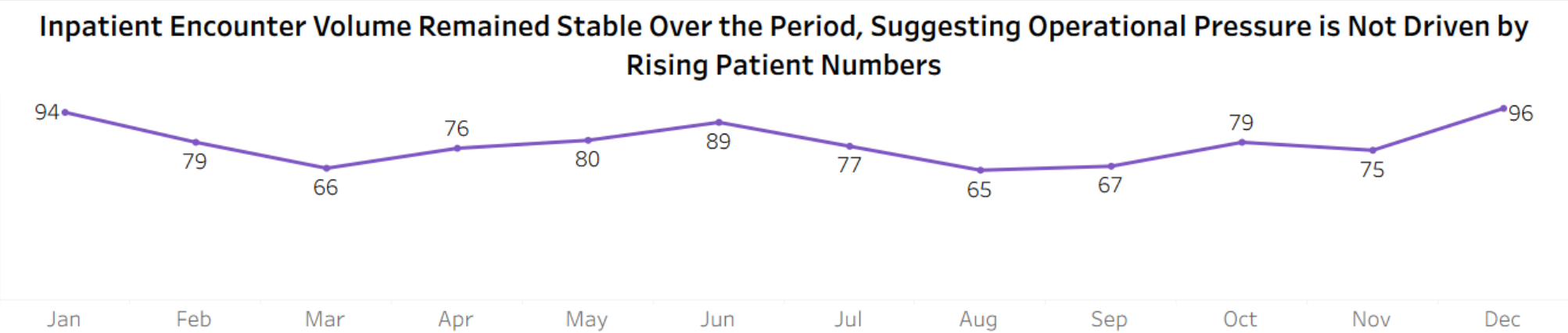
1,907

Total Bed Day Burden

9,061

Readmission Rate

9.9



OPERATIONAL DIAGNOSTICS: VOLUME VS. CAPACITY PRESSURES

CORE INSIGHTS & DATA TRENDS:

SEASONAL INPATIENT VOLUME PATTERNS

The top trend line tracks total encounter volume over time, representing the distinct number of individual inpatient admissions and clinical care processes managed by the hospital.

Data shows predictable winter surges, peaking at 94 encounters in January and 96 encounters in December, which perfectly aligns with end-of-year clinical trends.

Outside of seasonal winter surges, baseline inpatient volume is highly stable. This proves operational strain is not driven by an influx of new patient numbers.

OPERATIONAL DIAGNOSTICS: VOLUME VS. CAPACITY PRESSURES

PERSISTENT OCCUPANCY PRESSURE

The bottom trend line shows that Average Length of Stay (ALOS) remains elevated year-round, frequently peaking between 5.1 and 5.6 days (March, May, August, October) completely independent of volume dips.

PROLONGED INPATIENT STAYS

Sustained high ALOS validates concerns regarding delayed patient transition. Because stays remain high during low-volume months, the root cause is isolated to internal operational bottlenecks.

OPERATIONAL IMPLICATIONS:

Workload Bottlenecks:

1,907 total encounters consumed 9,061 bed days, straining ward resources.

Bed Gridlock Risks: High average stays block ward capacity and stall new inbound patient transfers.

Operational Analytics

Average LOS

4.7

Inpatient Volume

1,907

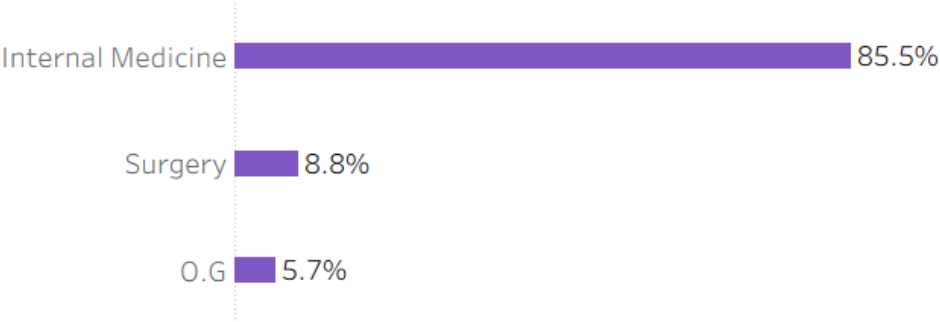
Total Bed Day Burden

9,061

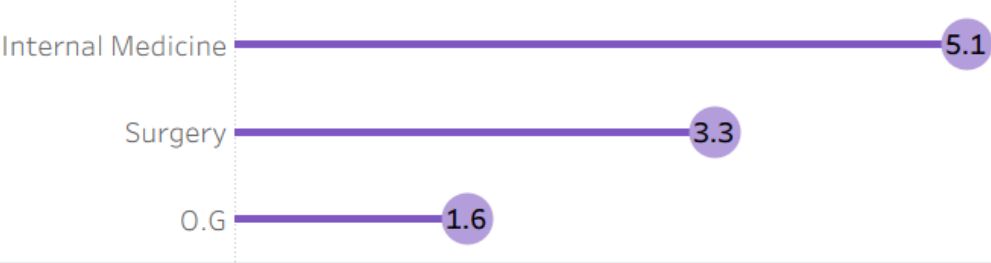
Readmission Rate

9.9

Internal Medicine Drives 86% of Inpatient Volume



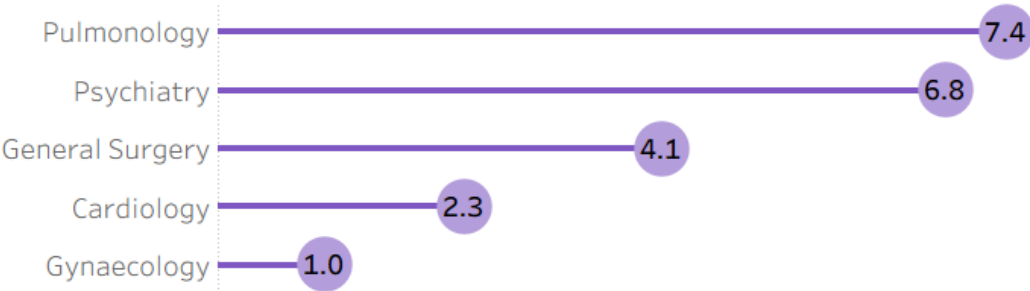
Internal Medicine Drives the Longest Patient Stays, Averaging 5.1 Days



Cardiology, Psychiatry, and Pulmonology Dominate Inpatient Specialty Volume



Pulmonology Records the Longest Average Stay Despite Ranking Third in Volume



DEPARTMENTAL DIAGNOSTICS: CAPACITY BOTTLENECK

CORE INSIGHTS & DATA TRENDS:

Internal Medicine Dominance: Accounts for 85.5% of total volume and the highest department ALOS of 5.1 days, indicating high centralized pressure.

Specialty Stay Outliers: Pulmonology (ALOS: 7.4 days) and Psychiatry (ALOS: 6.8 days) consume a disproportionate share of bed capacity.

IMPACT ON HOSPITAL OPERATIONS:

Patients in Pulmonology and Psychiatry utilize beds nearly twice as long as other units, creating a severe bed block that directly causes downstream admission bottlenecks.

Clinical Analytics

Average LOS

4.7

Total Inpatient Discharge

1,917

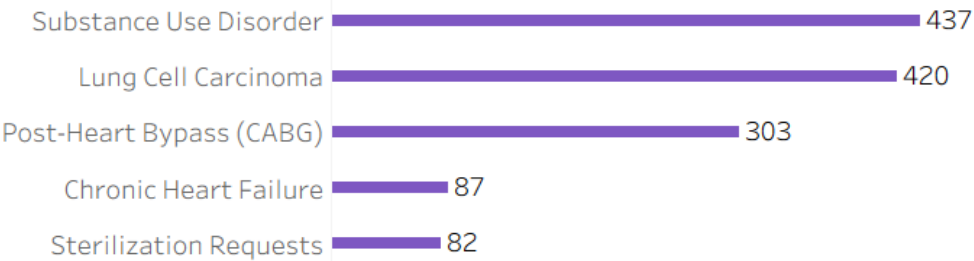
Total Readmission

189

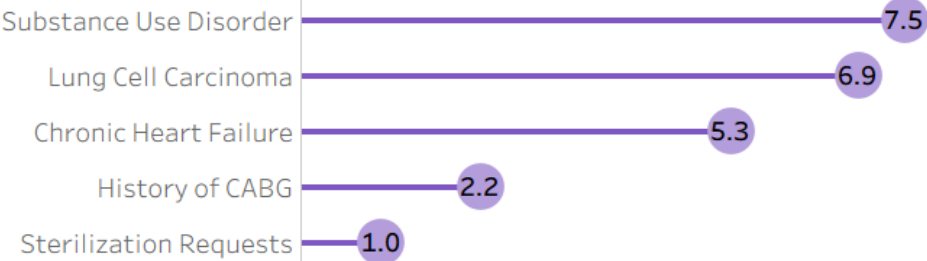
Readmission Rate

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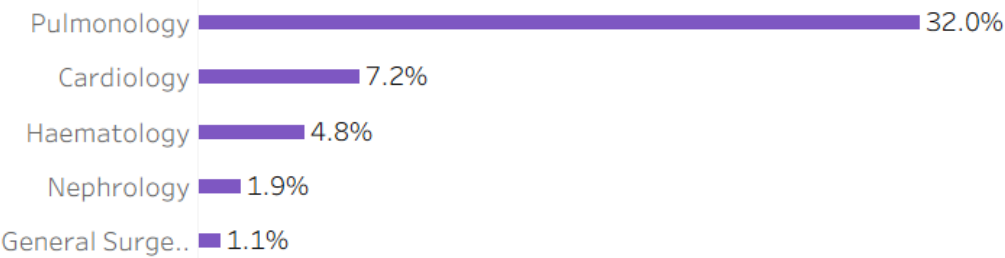
Substance Use Disorder and Lung Cancer are Consuming the Majority of Inpatient Capacity



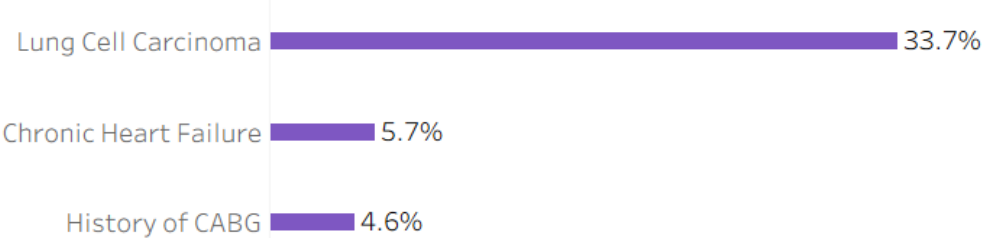
Substance Use Disorder and Lung Cancer are Also the Longest Staying, Creating Double Burden on Inpatient Capacity



Pulmonology Drives a Critical System-Wide Readmission Bottleneck



One in Three Lung Cell Carcinoma Patients Are Readmitted Post-Discharge



CLINICAL ANALYTICS: DIAGNOSIS & READMISSION DRIVERS

CORE INSIGHTS & DIAGNOSIS DRIVERS:

Substance Use & Lung Cancer: Drills down Psychiatry and Pulmonology's volume. Stays average 7.5 and 6.9 days respectively, creating high capacity demands.

The Readmission Crisis: Pulmonology accounts for 32% of total readmissions. Over 1 in 3 (33.7%) Lung Cancer patients are readmitted post-discharge.

IMPACT ON HOSPITAL OPERATIONS:

While Substance Use causes long initial stays, Lung Cancer creates a repeating cycle of re-hospitalization, severely locking up inpatient beds.

Financial Analytics

Insurance Payment
Patient Payments

Gross Billed Amount

495.8M

Total Net Revenue

495.8M

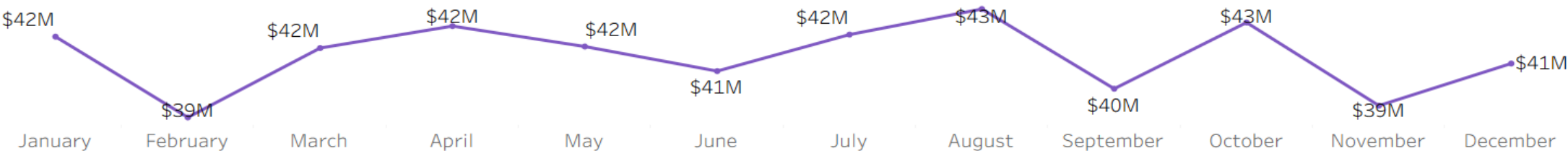
Patient Contribution
Rate

19.1%

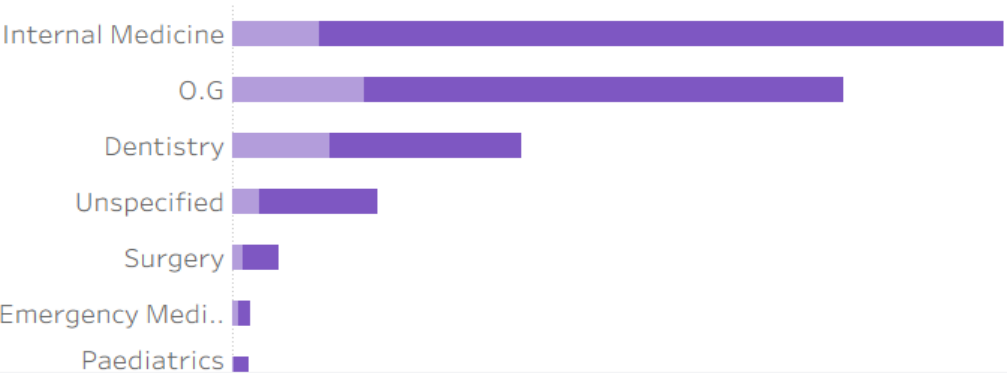
Average LOS

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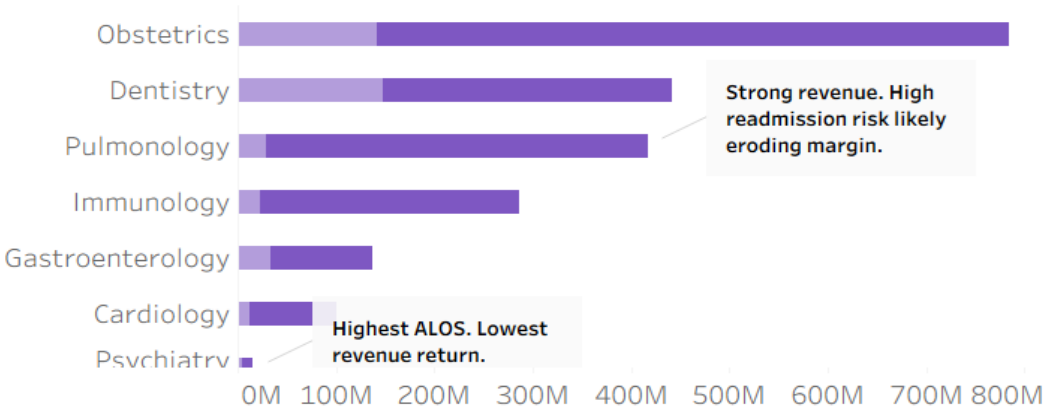
Combined Monthly Revenue Has Plateaued Between \$41M and \$43M With No Meaningful Growth Over the Period



Internal Medicine and O.G Drive the Majority of Departmental Revenue
But Financial Strain Persists Across the Hospital



Revenue Does Not Reflect Operational Burden Across Key Specialties



FINANCIAL ANALYTICS: CAPACITY CONSTRAINTS COST

FINANCIAL CONSEQUENCES OF SYSTEM DELAYS:

Stagnant Revenue: Monthly revenues plateaued at \$41M-\$43M due to fixed capacity ceilings. When beds are full, the hospital cannot admit new patients.

Disproportionate Yields: Psychiatry consumes huge bed day resources but generates the lowest relative yield. Margins are eroded by long stays.

KEY FINANCIAL DIAGNOSTIC:

High-burden clinical specialties with extremely high average stays limit hospital turnaround. Resolving clinical delays directly unlocks new patient admission pathways.

RECOMMENDATIONS: OPERATIONAL LEADERSHIP

IMMEDIATE ACTIONS

- **Deploy Focused Resource Staffing:** Increase nursing and operational staff support in Pulmonology and Psychiatry to coordinate timely workflows.
- **Implement post-discharge follow-up protocols :** Schedule phone calls or clinic follow-ups within 48-72 hours of discharge to catch early warning signs before readmission becomes necessary.
- **Strengthen discharge planning for Pulmonology:** Develop protocols specifically for high-readmission diagnoses like Lung Cancer to ensure patients leave with clear post-discharge instructions.

FURTHER INVESTIGATION

- **Audit Inpatient Care Paths:** Map the post-discharge care journey to target community medical services and limit readmission.

RECOMMENDATIONS: CLINICAL LEADERSHIP

IMMEDIATE ACTIONS

- **Post-Discharge Care Programs:** Prioritize follow-up clinics and community-based telehealth for high-risk Lung Cancer cases.
- **Develop disease-specific pathways:** Create targeted care pathways for Lung Cancer and Substance Use cases, addressing the specific reasons these patients return.
- **Deploy early warning systems:** Identify patients at high readmission risk BEFORE discharge so interventions can happen earlier in their care journey.

CLINICAL AUDITS

- **Discharge Readiness Audit:** Audit clinical discharge timelines versus administrative blocks to isolate delays.
- **Pulmonology Readmission Audit:** Target oncology treatment complications causing high readmit loops.
- **Psychiatric Transition Review:** Assess long-stay cases for step-down behavioral placement gaps.

RECOMMENDATIONS: FINANCIAL LEADERSHIP

STRATEGIC RESOURCE ALLOCATION

- **Invest in High-Burden Service Lines:** Fund clinical transition teams specifically in Pulmonology and Psychiatry to limit avoidable stays.
- **Dynamic Budget Allocation:** Adapt department budgets to match actual bed-day demands and capacity requirements.

FINANCIAL QUANTIFICATIONS

- **Determine Margin of Stays:** Calculate daily costs of ward occupancy in over-utilized units to establish actual loss margins.
- **Quantify Readmission Cost Burden:** Map systemic financial penalties of high-readmit patients on current insurance plans.

DATA LIMITATIONS & SCOPE CONSTRAINTS

ABSENCE OF DIRECT COST AND MARGIN METRICS

The dataset lists Gross Billed Amount and Net Revenue, but lacks internal operational cost accounting metrics. We cannot calculate exact margins or financial loss per extended stay.

LACK OF CLINICAL SEVERITY & ACUITY DATA

No clinical acuity tracking scores exist in the database. We cannot definitively say if a 7.5-day stay is driven by workflow delays or necessary clinical stabilization.

NO POST-DISCHARGE PATIENT TRACKING

Outpatient medication adherence and clinical follow-ups are untracked. The root causes of the 33.7% Pulmonology readmissions cannot be definitively mapped to external gaps.